

Chengze(Ritz) Sun

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EDUCATION

University of Waterloo

Candidate for BAsC. in Computer Engineering;

Waterloo, Ontario

Sept 2023 – May 2028

University of California, Berkeley

Study in Computer Science and Astronomy;

California, United States

Jun 2022 – Aug 2022

SKILLS

Languages: C/C++, C#, Java, Python, Swift, Lua

GPU/Systems: UE5, CUDA, Vulkan, OpenGL, GLSL/SPIR-V, GPU Memory Management, Synchronization

ML/Compute: PyTorch, NumPy

Frameworks/Libraries: Nsight Compute, Nsight Systems, RenderDoc, XCode, Linux, Git, CMake

WORK EXPERIENCE

Gameplay/Engine Engineer [Hoyoverse(miHoYo)] | *Co-op*

May 2026 – Aug 2026

- Built gameplay telemetry and analytics tools for an unannounced game project in pre-production UE5 tps competitive MOBA, including match-pacing analysis, kill heatmaps, and ability-selection analysis, enabling designers to analyze player behavior and rapidly tune match flow.
- Built an editor-side LAN multiplayer workflow in UE5, developing Slate tooling for creating and discovering UDP-based lobby servers across local editor instances, with support for PIE and standalone clients as well as in-process and standalone servers.
- Extended the packaged game with a LAN login workflow alongside the existing online authentication path, modifying Blueprint login interfaces and implementing a dedicated LAN lobby flow while reusing the underlying lobby and networking systems across editor and packaged builds.
- Contributed to Agent-assisted engineering infrastructure, including automated changelist code review and daily build performance analysis pipelines.
- Drove end-to-end testing and workflow validation for an internal Cursor-like AI production platform spanning code assistance, 3D content creation, and AI-driven animation generation.

Gameplay/Graphics Engineer [Digital Extremes] | *Co-op*

Jan 2026 – Apr 2026

- Contributed to gameplay and evolution engine systems for *Warframe*, implementing gameplay mechanics and runtime systems in C++ within the proprietary engine.
- Implemented dialogue camera systems supporting dynamic camera transitions and interpolation, integrating runtime view updates with the engine's rendering pipeline.
- Contributed to the *Warframe: The Shadowgrapher* update, implementing gameplay features and participating in gameplay system reworks across multiple modules.

Graphics Software Engineer [CoreAVI (Lynx)] | *Co-op*

May 2025 – Aug 2025

- Investigated and resolved failures and hangs in Vulkan CTS, adding debugging instrumentation and reducing the number of non-passing tests on internal driver branches.
- Analyzed validation errors and GPU crashes to track down incorrect synchronization, descriptor usage, and pipeline setup in driver and test code.
- Helped build a graphics compositor demo that rendered Vulkan SC, OpenGL SC, and OpenGL windows simultaneously, demonstrating multi-API interoperability on safety-critical platforms.

GPU Driver Engineer [Huawei Technologies Co., Ltd. Canada] | *Co-op*

Sept 2024 – Dec 2024

- Assisted the GPU driver team in evaluating new rendering technologies such as D3D12 work graphs, summarizing their implications for future driver design.
- Implemented and tuned shaders for internal demos, measuring performance using vendor profiling tools and iterating on bottlenecks.
- Fixed rendering artifacts and correctness issues by debugging command buffers, resource layouts, and synchronization in Vulkan/OpenGL paths.